

Week 4 — Task 10: What is SIEM? + Install Splunk / Wazuh

Status: Completed SIEM: Splunk

1. Project Overview

This project covers the fundamentals of Security Information and Event Management (SIEM) and demonstrates access to a working Splunk-based SIEM environment. A SIEM centralizes security-relevant logs and events, normalizes information, correlates related activity, supports detection and alerting, and provides dashboards for investigation.

2. Objectives

- Understand what SIEM means.
- Understand centralized log collection, normalization and correlation.
- Understand alerting and dashboards.
- Complete the TryHackMe Introduction to SIEM room.
- Access and verify a working SIEM dashboard.

3. What is SIEM?

SIEM stands for Security Information and Event Management. It collects logs from different sources, standardizes data, correlates events, detects potentially malicious activity, generates alerts, and presents information through search and dashboards.

4. Core SIEM Functions

Centralized Log Collection

Gathers logs from endpoints, servers, firewalls, network devices and applications into one platform.

Log Normalization

Converts different log formats and fields into a consistent structure.

Log Correlation

Connects related events from different sources so activity can be investigated as a sequence.

Detection and Alerting

Detection rules identify suspicious conditions and can generate alerts for analyst investigation.

Dashboards and Reporting

Summarizes security activity through event counts, alerts, trends and other metrics.

5. Three Benefits of SIEM

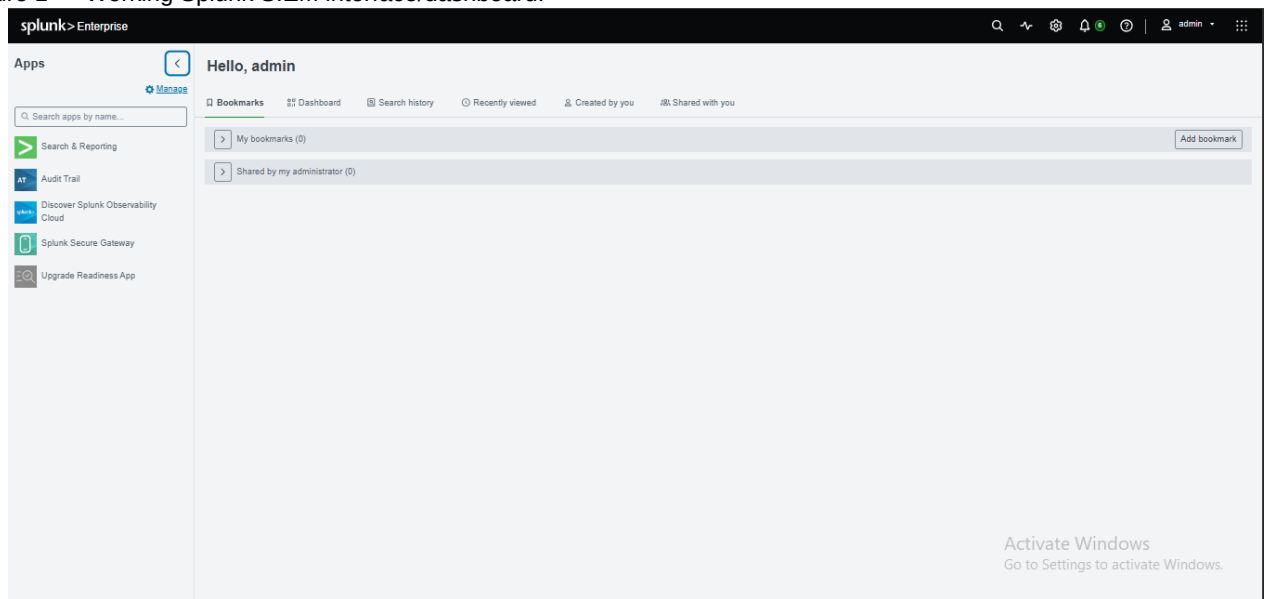
- Centralized visibility — security events can be investigated from one platform.
- Faster detection — correlation and detection rules help identify suspicious patterns efficiently.
- Better investigation and reporting — searchable events and dashboards provide evidence for response and escalation.

6. Practical SIEM Used

Splunk was used as the practical SIEM platform. An existing local Splunk environment was already installed and configured with security telemetry. Because system resources were limited, Wazuh was not installed alongside it. The existing Splunk setup was reused without deleting or rebuilding it.

7. Working Splunk Evidence

Figure 1 — Working Splunk SIEM interface/dashboard.



8. TryHackMe — Introduction to SIEM

The required TryHackMe Introduction to SIEM room was completed. The learning activity covered SIEM fundamentals including log sources, centralized collection, normalization, correlation, alerting, dashboards, ingestion and investigation.

Figure 2 — TryHackMe Introduction to SIEM evidence.

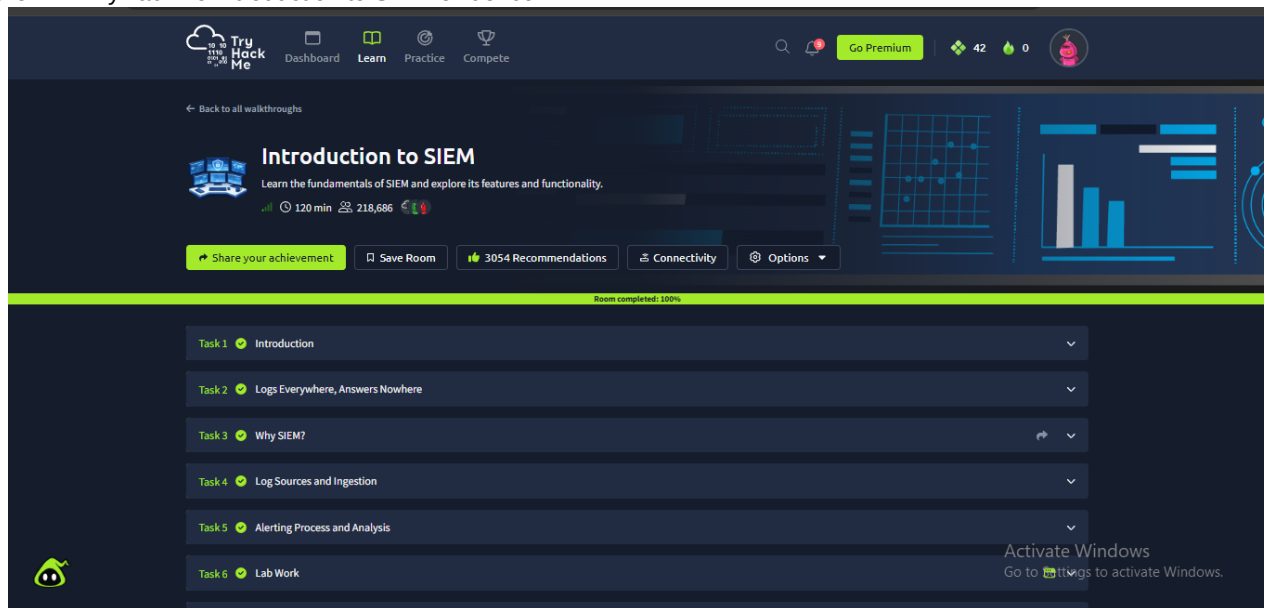
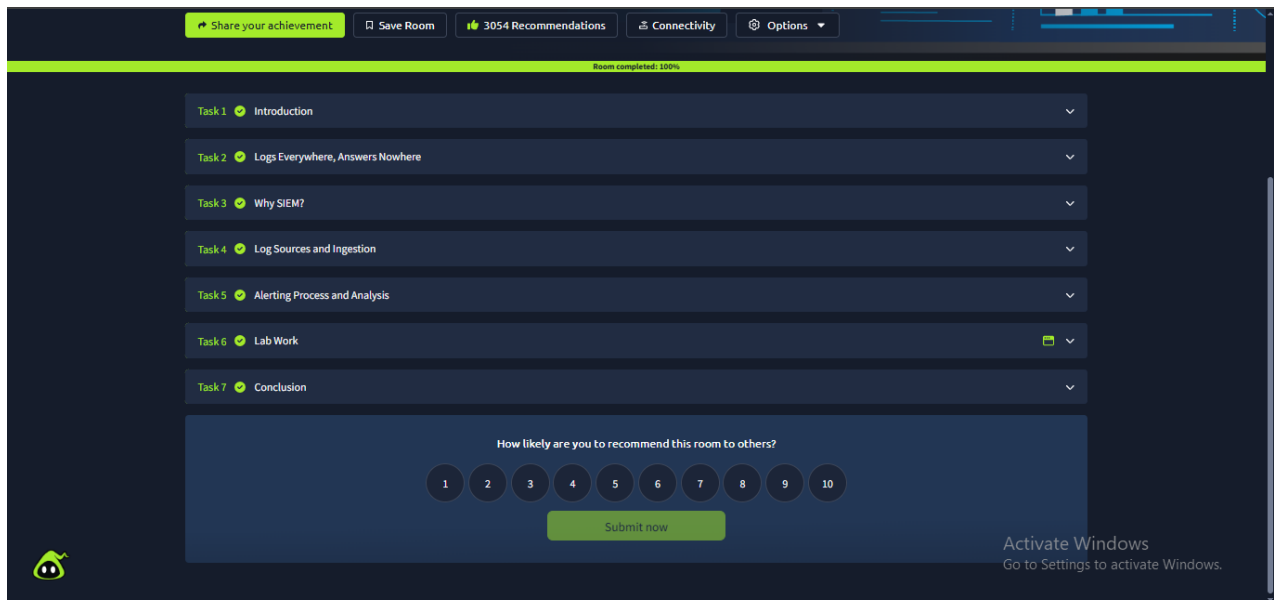


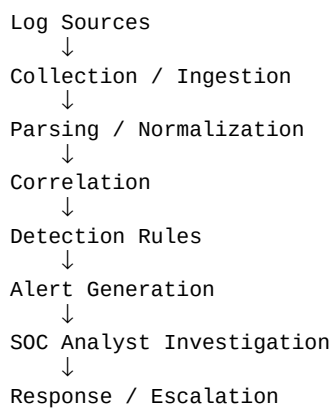
Figure 3 — TryHackMe Introduction to SIEM completion evidence.



9. Setup / Verification Steps

- Opened the existing Splunk environment.
- Confirmed that the Splunk web interface was reachable.
- Logged into the configured administrator interface.
- Verified that the dashboard/home interface loaded successfully.
- Completed the TryHackMe learning activity.
- Captured screenshots as evidence.

10. SIEM Workflow



11. SOC Analyst Relevance

A SOC analyst can use a SIEM to investigate failed authentication attempts, suspicious process execution, privilege escalation, persistence, suspicious DNS activity, unusual network connections, malware indicators and possible data exfiltration. SIEM evidence can be correlated with endpoint, firewall, DNS, authentication and other telemetry.

12. Practical Skills Demonstrated

Skill	Demonstration
SIEM Fundamentals	Explained SIEM purpose and capabilities
Log Collection	Understood centralized ingestion
Normalization	Understood consistent event fields
Correlation	Understood relationships between events
Alerting	Understood detection-triggered alerts

Skill	Demonstration
Dashboard Usage	Accessed working Splunk interface
Documentation	Produced structured evidence and report

13. Evidence Summary

- w1.png — Working Splunk SIEM interface/dashboard.
- w2.png — TryHackMe Introduction to SIEM evidence.
- w3.png — TryHackMe Introduction to SIEM completion evidence.

14. Repository Structure

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Week-4-Task-10-What-is-SIEM-Install-Splunk/
■
■■■ README.md
■■■ Week-4-Task-10-What-is-SIEM-Install-Splunk.pdf
■
■■■ Report/
    ■■■ report.md
    ■■■ w1.png
    ■■■ w2.png
    ■■■ w3.png

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15. Conclusion

This project established the fundamentals of SIEM and demonstrated a working Splunk environment. The practical work covered centralized collection, normalization, correlation, detection, alerting, dashboards and SOC investigation while reusing an existing lab under resource constraints.